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PROMOTING INNOVATION IN TEACHING AND LEARNING: PROBLEMS, PROCESSES AND INSTITUTIONAL MECHANISMS*

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ABSTRACT

This paper examines constraints on innovation in teaching and learning in higher education, and institutional mechanisms for trying to promote such innovation. Innovation is considered as a process of change rather than the dissemination of novel ideas. Factors such as teaching climate, resource allocation and consultancy support are considered. The dangers of a mismatch between “innovations” and problems are discussed, and the importance of giving adequate attention to problem diagnosis is stressed. A possible model incorporating these emphases is outlined in brief.

Introduction

The needs, goals and resources of institutions of higher education vary widely, and we should expect this to be reflected in their curricula and in their teaching patterns. Even within a single institution there is considerable variation. Differences in subject matter, differences in tradition and differences in personnel all make it probable that many problems of teaching and learning will be specific rather than general. Indeed what is an innovation in one context may well be traditional practice in another. Hence we prefer to define an innovation as a planned change in response to perceived problems rather than as the introduction of some new method or technique. It is the process of innovation which is critical rather than the content; and it is by examining problems and processes rather than new methods and techniques that we hope to analyse institutional mechanisms for the promotion of innovation in teaching and learning. This paper attempts to answer three questions. What are the

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problems of innovation in teaching and learning? What kind of policy should an institution have for promoting such innovation? What, if any, should be the role of the various support services?

Constraints on Innovation

Let us start by examining some of the constraints on innovations in teaching and learning which form part of the general context of higher education. All the following constraints are significant barriers to innovation in higher education and strategies which seek to ignore them (or banish them by wishful thinking) are unlikely to yield long-term benefits.

No dissatisfaction with the present solution

Unless a teacher perceives some discrepancy between his goals and his achievements, he is bound to regard innovation as undesirable and unnecessary; and the extent to which his expressed dissatisfaction is fundamental or trivial will determine the extent to which he is likely to entertain innovation. If a teacher does not have a problem, innovation will seem irrelevant. Dissatisfaction is most likely to arise from his criticism of his goals, from his assessment of student course-work and examinations or from his perceptions of the climate of opinion about his teaching, both faculty opinion and student opinion. It is less likely to arise from being made aware of “better” alternatives as there is no perceived need for anything “better”. In this context the “teaching climate” of the institution is of considerable importance, and the presence or absence of support services relatively unimportant, except insofar as they influence that climate.

No priority given to teaching and learning

It is a commonplace belief that the teaching role is considered relatively unimportant by many members of faculty and almost totally ignored when making appointments and promotions. When this is actually the case, the rewards of innovation are wholly intrinsic; and the common perception of innovation in such circumstances is that it means spending significantly *more* time on teaching and hence *less* time on research and administration.

Few resources are directed specifically to innovations

The existence of support services such as the “high activity” centres for educational technology supported by the University Grants Committee in the U.K. could be construed as a sign that some resources are

being directed specifically to innovation. But the provision of educational technologists for this purpose is not sufficient. Far more faculty time is needed than educational technologist time. Since it is likely that faculty will only rarely be willing to spend significantly more time on teaching, time for innovation has to come from a reallocation of existing teaching resources.

Little guidance on how to develop profitable innovations

A teacher needs to solve *his own* problems on *his own* terms, but often has very little idea of how to proceed. Many teachers have wasted time on relatively unsuccessful innovations because they have lacked appropriate guidance; often they have never adequately defined the real nature of their problems. These abortive attempts at innovations have helped promote a climate of opinion which believes that little improvement is possible and all would-be innovators are “false prophets”. Many regard these “inoculations” against innovation as necessary maturation experiences for university teachers.

Ignorance about possible innovations

There is undoubtedly considerable ignorance about new methods of teaching and learning, about the research literature on new and old methods, and about innovations in other universities and in other parts of the same university. This is certainly a barrier to innovation though its significance is possibly overrated. Is there any point, for example, in disseminating information to people who remain perfectly satisfied with the existing situation?

From this discussion of the problems of innovation we can conclude that there are three factors which affect the likelihood of successful innovation in teaching and learning taking place:

- (i) The teaching climate of the institution, i.e. the attitudes of faculty, students and administrators.
- (ii) The allocation of resources specifically for innovation, particularly the provision of release time for faculty to develop innovations.
- (iii) Provision of appropriate support services.

Changing the Teaching Climate

The teaching climate of an institution is difficult to describe and even more difficult to change. But it is worth examining some of the

factors which must inevitably affect it. The first is the institution's policy, or rather its normative practice, for appointments and promotions (Jones, 1973). Seen at its worst, it is not uncommon in universities for teaching either as a responsibility or as an area of concern to be omitted from all advertisements, interviews and discussions. It would be both impracticable and undesirable to try to attach any specific weighting to reported or even observed teaching ability, but to give one of the members of the committee concerned a specific responsibility for raising the issue would not seem unreasonable.

A second factor is the extent to which teaching and learning are a normal part of ongoing departmental and university discussions about problems and issues. This is the responsibility of the professor or dean, who not only can see that teaching is discussed but also can set the whole tone of the discussion. But even when he is willing to fulfil this role the task of ensuring a productive rather than a trivial or recriminatory discussion is not easy and often goes by default. It is an area where advice from an experienced consultant could be helpful, but we shall return to this later. There is no use just discussing a problem unless the discussants have some notion as to what appropriate actions might be possible and some conviction that decisions will be taken seriously and properly implemented.

The third factor, which is of growing importance, is the participation of students in discussions on teaching and learning. Student involvement can certainly increase the chances of teaching coming onto the agenda and of discussion leading to action, but it does not of itself ensure that the discussion will be productive. Mere complaints about teaching usually give rise to defensive reactions; and it requires a certain degree of mutual trust between faculty and students before problems of both teaching *and* learning can be discussed together. The teaching climate is both influenced by and has an influence on the general institutional climate and the nature of the relationships between students and the faculty.

Then finally the climate can be affected by some clear indication from the administration that it attaches importance to the teaching function. Firstly it has to build this attitude into the committee structure; and we shall be discussing the role of teaching and learning committees later. And secondly it has to provide resources for the development and improvement of teaching and learning.

Allocation of Resources for Teaching and Learning

Until very recently the allocation of resources specifically for inno-

vation in teaching and learning was confined to the provision of support services. They were principally in the area of educational technology; and they were justified by the potential for cost-reduction and by appealing to modernity. This policy reflected the views of the 1960s but experience has shown that it has a number of serious disadvantages:

(1) It has very little effect on the “teaching climate” of the institution.

(2) It tends to transfer the responsibility for innovation away from the faculty themselves and onto the support services, who may often find themselves being ignored and blamed for lack of impact at the same time.

(3) It underestimates the amount of extra work involved for members of faculty.

(4) Since much of their concern with teaching is inevitably “second hand,” the links between support services and students can be very tenuous indeed.

(5) It creates a situation in which there are “solutions” looking for problems, rather than the reverse.

(6) The pattern of innovation is inevitably random and spasmodic and there is little guarantee of it having any relevance to the main problems and needs of the institution.

A number of these disadvantages can be overcome if a wider range of support services are encouraged (we shall return to this later) and if at least equivalent funds are allocated specifically for faculty release time. It is not realistic to expect a member of faculty to sacrifice his research but it is possible to reallocate teaching resources so that this extra time for the development of innovations in teaching and learning comes at the expense of a slight worsening in the effective teacher-student ratio (Jones, 1973). After all, the evidence on the Hawthorne effect, which is the least one can expect from innovation, is a good deal more conclusive than that on class size! It is also possible, particularly in the sciences, to employ a research fellow or research assistant who might both undertake some teaching and maintain, or even strengthen, a group’s research output.

The provision of release time could also have a significant effect on the climate of the institution. It gives official recognition to innovation in teaching, and is also likely to encourage publication of the work and hence confer extra status upon it. This depends, of course, on the time being used in a properly professional way; and for someone both inexperienced and untrained for this kind of activity there can be no guarantee of this. Again the role of the support services could be important. There is a danger, however, of creating a separate, and inevitably lower status class of academic who studies the teaching problems while the others get on

with their research with a free conscience. To avoid this danger it is essential that the released person involves and consults his colleagues throughout rather than tries to make it on his own, even if it is at the expense of "productivity".

This highlights the fundamental test of an innovation strategy. Does it effectively locate the responsibility for innovation with the main body of faculty; or does it relieve them of the "burden" by locating the responsibility either with support services or with some non-representative group of faculty, who have been "foolish" enough to risk their careers by applying their talents to problems of teaching and learning? The provision of release time is a necessary but not a sufficient condition for passing this test.

Provision of Support Services

Three distinctive functions for support services are now discernible in current practice: the improvement of teaching skills, the development of resources for learning, and assistance with the design and evaluation of courses. The first of these, the improvement of teaching skills, is concerned with the teacher in his actor/presenter role. Traditionally this took two forms: exhortation to use visual aids and assistance with their production; and provision of checklists of "dos" and "don'ts" for lecturing, tutoring, discussion leading, etc. These tasks are by no means redundant although the strategy of performing them by lecturing or by circulating printed documents has proved markedly unsuccessful. There is evidence that visual aids are still ignored by large numbers of teachers and that many of the most elementary performance rules are broken every day. The reason, one suspects, is not lack of information but lack of motivation; and for that we must look not to the support services but to the teaching climate.

In any case the "inform and exhort" approach to the improvement of teaching skills has been superseded technologically by the use of portable closed-circuit television. Now instead of being lectured about the inefficiency of lectures, you can see them on television or even record your own. This has been refined still further in the approach called micro-teaching in which a teacher can practice a few minutes teaching on a small group of students, hear his recorded performance, try again with another group of students and so on, usually concentrating on one specific teaching skill at a time, e.g. questioning (Allen & Ryan, 1969). As normally practised this approach has two major drawbacks: it tends to concentrate on technical skills and to ignore the semantic and subject-specific

elements in teaching; and it sets up the consultant as the definer of what constitutes “good” teaching rather than another teacher or a student, thus isolating him from his clients and forcing him into the role of expert. Both these disadvantages have the effect of separating off general educational expertise from subject-specific expertise and make it easier to reject the consultant as not knowing the subject. Our own experience leads us to believe that a crucial factor in innovation is the complementing of general pedagogic expertise and experience with that possessed by good subject teachers; and that it is neither strategically desirable nor even theoretically valid to expect a general consultant to solve subject-specific problems on his own. One way round these difficulties is to try and involve groups of teachers in observing and helping each other; and also, if possible, to get students to say what *they* thought was going on. But this is more demanding of time and effort.

Undoubtedly the recording and review of teaching performance can contribute to the improvement of teaching skills and is an important function for support services to perform. But nevertheless it has yet to be seen whether the greater glamour of this quasi-therapeutic experience will entice more than a minority of teachers.

Those who lacked faith in “changing human nature” looked to improvements of a more technological kind and set up support services to develop alternative resources for learning (MacKenzie et al., 1970). Some who talked of giving the exceptional teacher a wider audience and of cutting costs have emphasised the potential of mass media, particularly television; and others who talked of changing the emphasis from teaching to learning and of adapting to the needs of individual students have emphasised first programmed learning and then other forms of independent study, such as the audiotutorial approach (Postlethwait, et al., 1969), the Keller plan (Elton et al., 1973), and modular instruction (Goldschmid and Goldschmid, 1973), to list them in historical order. But experience with both approaches has led to similar conclusions.

(1) The faculty time required in developing new resources is considerable and unlikely to be donated by all but a few enthusiasts unless there is provision for release time (see above).

(2) The development of new resources is far more profitable when pursued as part of the design or redesign of a whole course.

(3) Improvements are much more likely to be qualitative than quantitative.

(4) The teacher’s role is changed but not diminished. He is more likely to function as a tutor and counsellor and less likely to function as an actor/presenter.

More recently this kind of work has been given enormous impetus by the Open University which has been able to invest in learning resources on an appropriate scale, but we have yet to see whether in the end conventional universities in the United Kingdom will move more towards the Open University pattern or whether the Open University itself will become more conventional as the traditional academic pressures on its faculty grow with time. It is, however, our belief that the potential of new resources for learning is only beginning to be explored. Materials of quality may as yet be exceedingly scarce, but there could be many more in the future if the investment is made and if the proponents of alternative learning resources survive the double hazard of the sceptics and the enthusiasts.

“Teaching Methods Units” were originally set up to provide the first support function, the improvement of teaching skills; and “Educational Technology Units” were set up to provide the second, the development of learning resources. Both have tended to overlap into each other’s territory, often because only one of them actually existed in the institution concerned; and both have gradually assumed the third support function, that of assistance with the design and evaluation of courses (Beard, 1970; MacKenzie, et al., 1970). This third support area is much more controversial than the other two because it encroaches on traditional academic preserves and is therefore far more threatening. It also lacks an accepted body of educational knowledge which might make the consultant’s claim to be useful more convincing; and it relies even more than the other functions on the allocation of substantial faculty time to planning and evaluation. Why then have the support services chosen to invade such difficult and potentially unrewarding territory? Because to ignore it would make a nonsense of their other activities. What, after all, is the purpose of helping someone to improve a lecture when the students might be learning better from a book; and how can one design and evaluate resources for learning without first clarifying the aims of the course? The selection of appropriate methods must surely be as important as the improvement of them, but the change, being more radical, is far more difficult to accomplish.

Further Difficulties

An innovation strategy which was aimed at all three of these critical factors – the teaching climate, faculty release time and support services – was being publicly advocated by 1970; and by now it would probably receive widespread support from those engaged in the support services,

and even from official bodies such as the University Grants Committee (1972). In spite of still being very far from the normative practice it must be close to becoming the “conventional wisdom.” Nevertheless it does have serious weaknesses, some of which we have already hinted at. These include the relative isolation of support services from teachers and students, the difficulty of evaluating an innovation’s success, the scanty attention paid to problem diagnosis and the tendency to concentrate on problems which conform to readily available solution paradigms. We shall consider each of these problems in turn before proceeding to consider a modified strategy which attempts to take them into account.

The support services will always be somewhat isolated from teachers and students because they serve a different function; but reducing the degree of isolation should be an important part of any innovation strategy. One approach is to plan for a greater overlap of functions, giving support service consultants at least a minor teaching role and appropriately skilled and experienced members of faculty a consultancy role. Another is to increase the level of informal contact between consultants and faculty and, if possible, students. But even a strong emphasis on maintaining contacts of all kinds and at all levels will be of little use if there is intellectual isolation. What is sometimes disparagingly referred to as a “jargon gap” can be symptomatic of something much deeper: differing views on both the nature and the validity of educational expertise. At one extreme there are those who regard educational knowledge as narrow in scope but high in validity; there are one or two ways of giving a good lecture or a good lesson and all you have to do is to learn them. Teacher training should consist solely of taking subject experts and giving them a crash course on teaching methods. Then at another extreme there are those who regard educational knowledge as wide in scope and high in validity. Such people usually see themselves as a cross between a behavioural psychologist and a systems engineer; and for them a member of faculty is solely a subject expert to be interviewed, never a teacher to be consulted. Indeed they find it inconceivable that teachers should have any important knowledge to contribute to the subject of teaching and learning. Then thirdly there are those who, like ourselves, regard educational knowledge as wide in scope but low in validity. As educational technologists we have come to support Galbraith’s statement that “technology means the systematic application of scientific and other organised knowledge to practical tasks” (Galbraith, 1968), but remain somewhat doubtful as to how much scientific and other organised knowledge actually exists in the field of education. Whereas the narrow scope, high validity view locates nearly all relevant expertise with the subject specialist and the wide scope, high validity view locates virtually all the expertise

with the consultant, we believe in the concept of complementary expertise. Although it is possible for subject specialists to solve teaching problems solely on the basis of their own knowledge and experience and that of their colleagues, it is also possible that they will benefit from the additional knowledge and experience of a consultant. This position of complementary expertise is quite difficult to maintain in practice, because the divide between a consultant who cannot solve problems on his own, without faculty participation and assistance, and a consultant who is no use at all can sometimes appear dangerously narrow.

The problem of evaluation is also a significant one, and again there are two extremes. One extreme, the tendency to regard the mere appearance of a change as an indicator of an improvement, is particularly tempting for support services as it makes it so much easier for them to justify their existence. The fact that a member of faculty has decided to show some films or consulted an educational technologist does not in itself indicate that any improvement in teaching has taken place. Any attempt to regard such activities as aims in themselves will tend to lead to a concentration on marginal changes which are easy to make at the expense of more fundamental innovations which take longer to achieve. This is particularly dangerous when refinement of the existing pattern of teaching and learning hinders the development of a new and more desirable pattern.

This same problem of marginal versus insignificant change arises with the opposite position: the demand for unambiguous quantitative indicators of success. It is only possible to get purely quantitative indicators of success if the following conditions apply:

- (a) the objectives of the course are unchanged.
- (b) the assessment pattern is unchanged.
- (c) there is no mismatch between the intentions of the faculty and the assessment.
- (d) the student intake is unchanged.
- (e) the course context is unchanged.

These conditions are rarely met in practice. But the alternative, which is to set up a controlled experiment, is not popular with faculty or students. Nor is it free from methodological problems, as we have indicated elsewhere (MacKenzie et al., 1970). The evaluation of an innovation must necessarily involve both quantitative and qualitative components as well as personal value judgements (Parlett, 1972; Stake, 1967; Trow, 1970). But there is very little evidence of this kind of evaluation being met in practice on any significant scale, partly one presumes because of lack of training and partly because of lack of resources.

The last two difficulties listed above are closely related: the scanty

attention paid to problem diagnosis; and the tendency to concentrate on problems which conform to readily available solution paradigms. When resources are scarce few people want to “waste” them on preliminary investigations which have no obvious and immediate pay-off. So the strategy we have been discussing inevitably leads to a concentration of the available resources on development work, i.e. problem solution, without pausing to consider whether it is the real problem that is being tackled or merely some of its more obvious symptoms. Since there is a similar reluctance to “squander” resources on evaluation a mistaken diagnosis of a problem can take a long time to become apparent. The tendency to concentrate on readily available paradigms for tackling teaching problems has the same origin, the urge to get on with the job. Moreover it is magnified by the fact that these paradigms often form part of the expertise which the consultant brings to the situation; and he also is tempted to seek out and define problems in such a way that his own knowledge and experience is clearly perceived as relevant, a tendency that is reinforced by the marginal and insecure nature of his job.

A Modified Strategy Incorporating a Problem-Oriented Approach

We stated earlier that innovation is unlikely to occur unless there is some awareness of a problem; but we omitted to point out that the obvious corollary of this axiom is false. When innovation does occur it is not necessarily related to the nature of the problem in any direct and logical way. As we implied when discussing the neglect of proper problem diagnosis, problem awareness can easily lead to hasty innovation without any useful investigation of the nature of the problem. This mismatch between innovations and problems is encouraged by their entirely separate treatment in the literature of higher education. If this paper had been primarily concerned with the content of innovation, especially with “new” methods and techniques, it would probably have included the following headings: microteaching, project work, continuous assessment, course development, group study, television, simulation, Keller plan courses, etc. But if we compare this list with a list of problems such as that drawn up below, there is no obvious one-to-one correspondence.

Student difficulties in working independently.

Lack of interaction in tutorials.

High drop-out rate.

Students feel formal workload gives no time for independent thought.

Students do not discuss work with each other.

Students perceive curriculum as “irrelevant” or “ideologically biased.”

Use of source material is insufficiently critical.

Examinations show lack of understanding of important concepts.

Insufficient attention to problems of individual students.

Lectures dull or difficult to understand.

Unnecessary peaking of demands on library for certain books.

First-year courses take insufficient account of different student knowledge.

Weaker students are harassed rather than educated.

Courses are unrelated to each other.

Discussion in seminars tends to be random and inconsequential.

Insufficient use of literature.

A list of innovations merely gives the latest additions to a continually expanding repertoire of ideas and techniques whose relevance to any particular problem is unlikely to become apparent until the nature of the problem has been fully explored; and even then one would not be surprised if most of the problems were tackled without involving any of the typical “conference list” of innovations. So instead of approaching consultancy situations with a pocket-book of innovations and a budget aimed solely at providing resources for development work, we are attempting to evolve a problem-oriented strategy using Fig. 1 as a model.

This model distinguishes five stages – Problem Awareness, Problem Diagnosis, Problem Study, Problem Attack and Evaluation – and two main types of activity – formal activities such as survey research, studying and contributing to the literature, committee proposals, training and liaison with other institutions; and less formally organised discussion and consultancy. Each stage, except possibly the first, might be delegated to a task group responsible both to the institution’s main teaching and learning committee and to the academic area involved. This group would contain faculty from the academic area concerned, students and consultants, the consultancy role being assumed by other members of academic faculty as well as people from the support services. It would also be able to use both the release time of members of faculty and the time of research assistants, provided that they had been appropriately anticipated in the budget. Thus some members of the task group would only have a discussant role but others would have a working role as well.

The total budget for this process would be under the control of the institution’s teaching and learning committee; and they would be responsible both for setting up the task groups and for deciding which problems should be given priority. In addition they would receive the annual reports

Formal activities

Discussion and consultancy

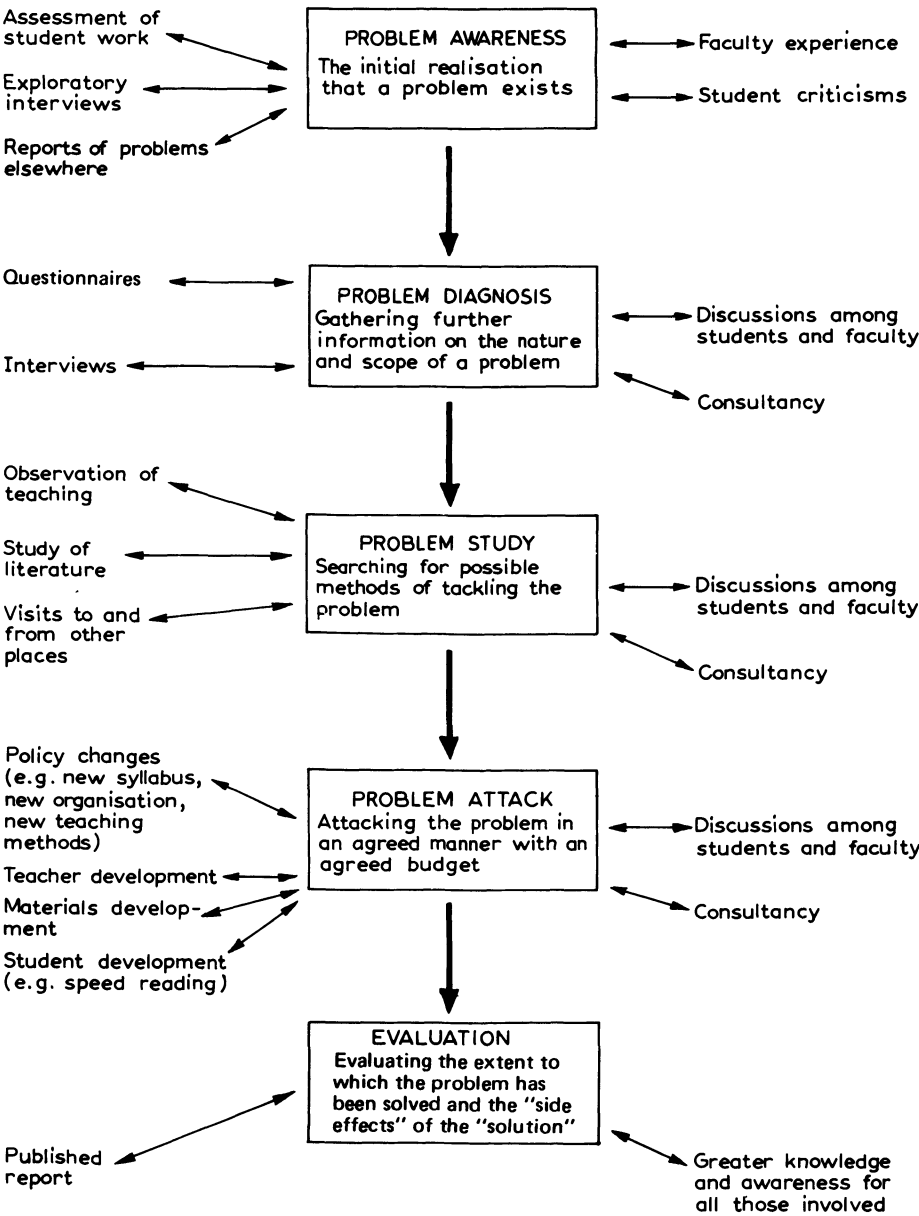


Fig. 1. A model for identifying and tackling problems of teaching and learning.

of support service units, and would be able to call on them for people to man the task groups. Other functions might include coordinating information, discussing those aspects of university policy which affected teaching and learning, and a special responsibility for stimulating problem awareness. This last function is probably the most difficult of all. Our own experience would suggest that it needs to be planned on a separate basis for each school, faculty or department; and that a mixed strategy might be useful. This could include both problem scanning, using the sources of information suggested at the top of Fig. 1 -- faculty experience, student criticisms, assessment results, exploratory interviews, and reports of problems elsewhere -- on a regular and organised basis; and the consideration of detailed reports of possibly relevant innovations in neighbouring departments or in similar departments in other institutions. But even this strategy will ultimately depend for its success on the teaching climate, which is where we first came in!

References

- Allen, D. W.; Ryan, K. A. (1969). *Microteaching*. Reading, Mass.: Addison-Wesley.
- Beard, R. (1970). *Teaching and Learning in Higher Education*. London: Penguin.
- Elton, L. R. B., et al. (1973). "Teach Yourself Paradigm -- the Keller Plan," *Chem. in Brit.* (April). 9: 164.
- Galbraith, J. K. (1968). *The New Industrial State*. London: Penguin.
- Goldschmid, B.; Goldschmid, M. L. (1973). "Modular Instruction in Higher Education: A Review," *Higher Educ.* 2: 15-32.
- Jones, H. C. (1973). "The Management of Teaching," in Fielden, J. and Lockwood, G., eds., *Planning and Management in Universities: A Study of British Universities*. London: Chatto and Windus, for Sussex UP.
- MacKenzie, N.; Eraut, M. R.; and Jones, H. C. (1970). *Teaching and Learning: New Methods and Resources in Higher Education*. Paris: Unesco/IAU.
- Parlett, M. (1972). "Evaluating Innovations in Teaching," in Butcher, H. J. and Rudd, E., eds., *Contemporary Problems in Higher Education*. London: McGraw-Hill.
- Postlethwait, S. N.; Novak, J. D.; and Murray, H. T. (1969). *The Audio-Tutorial Approach to Learning*. 2nd ed. Minneapolis: Burgess.
- Stake, R. E. (1967). "The Countenance of Educational Evaluation," *Teach. Coll. Rec.* (April). 68.7:533.
- Trow, M. (1970). "Methodological Problems in the Evaluation of Innovation," in Wittrock, M. C. and Wiley, D. E. eds., *The Evaluation of Instruction: Issues and Problems*. New York: Holt Rinehart.
- University Grants Committee (1972). *Annual Survey for Academic Year 1970-1*. Appendix IV. London: HMSO.